SQL Query Implementation for E-Commerce Database

OBJECTIVE

1. To perform SQL queries using `SELECT`, `WHERE`, `ORDER BY`, and `DISTINCT`.

2. To search products based on price, category, and availability.

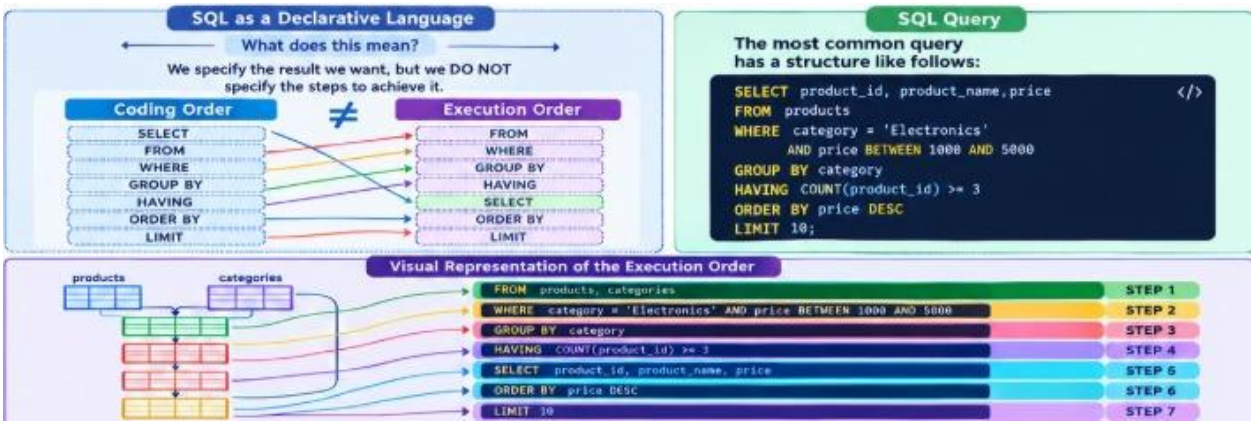
3. To retrieve customer and product information from the database.

4. To apply filtering conditions using `AND`, `OR`, `IN`, and `BETWEEN`.

5. To generate basic business reports, including product price reports, customer order reports, and total purchase reports.

1.Basic SQL Queries

1.1 SELECT Query



Display all products

```
SQL> SELECT * FROM PRODUCT;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY                                           PRICE      STOCK
-------------------------------------------------- ---------- ----------
         1
Laptop
Electronics                                         45000         10

         2
Shirt
Fashion                                               799         25

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY                                           PRICE      STOCK
-------------------------------------------------- ---------- ----------

         3
Headphones
Electronics                                          1500         15

         4
Shoes

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY                                           PRICE      STOCK
-------------------------------------------------- ---------- ----------
Footwear                                             1999         20

         5
Watch
Accessories                                          2500         12
```

Display selected product details

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE, STOCK
  2  FROM SHOP_PRODUCT;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       101
Wireless Headphones
      1799         30

       105
Electric Kettle
      1299         20

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

       106
Bluetooth Speaker
      1599         20

       107
Casual Shirt

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       799         35

       108
Face Wash
       499         15

       109
```



1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with price less than 1000

```
SQL> SELECT *
  2  FROM SHOP_PRODUCT
  3  WHERE PRICE < 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       107
Casual Shirt
        102        799         35

       108
Face Wash
        104        499         15

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

       103
Cotton Shirt
          2        899         40
```

Find products with price greater than 100

```
SQL> SELECT *
  2  FROM SHOP_PRODUCT
  3  WHERE PRICE > 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
      101
Wireless Headphones
         1       1799         30

      105
Electric Kettle
         3       1299         20

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

      106
Bluetooth Speaker
        101       1599         20

      109
Handbag
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
       102       1299         10

      110
Electric Kettle
       103       1199         18

      102

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
Smart Watch
         1       2999         15

      104
Denim Jeans
         2       1499         30


7 rows selected.
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from lowest to highest price

```
SQL> SELECT *
  2  FROM SHOP_PRODUCT
  3  ORDER BY PRICE ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
       108
Face Wash
       104        499         15

       107
Casual Shirt
       102        799         35

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------

       103
Cotton Shirt
         2        899         40

       110
Electric Kettle

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
---------- ---------- ----------
```

```
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID       PRICE       STOCK
----------- ---------- ----------
Handbag
         102        1299          10

       104
Denim Jeans
           2        1499          30


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID       PRICE       STOCK
----------- ---------- ----------
       106
Bluetooth Speaker
         101        1599          20

       101
Wireless Headphones
           1        1799          30

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID       PRICE       STOCK
----------- ---------- ----------

       102
Smart Watch
           1        2999          15


10 rows selected.
```

Products from highest to lowest price

```
SQL> SELECT *
  2  FROM SHOP_PRODUCT
  3  ORDER BY PRICE DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       102
Smart Watch
          1       2999         15

       101
Wireless Headphones
          1       1799         30

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

       106
Bluetooth Speaker
        101       1599         20

       104
Denim Jeans

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
Electric Kettle
          3       1299         20

       110
Electric Kettle
        103       1199         18


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       103
Cotton Shirt
          2        899         40

       107
Casual Shirt
        102        799         35

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

       108
Face Wash
        104        499         15


10 rows selected.
```

2.Search Products

2.1 Search Products Based on Price

```
SQL> SELECT *
  2  FROM SHOP_PRODUCT
  3  WHERE CATEGORY_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       901
Pen
          2        150         50

       902
Book
          2        250         30

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

       903
Pencil
          2        100         60

       103
Cotton Shirt

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
          2        899         40

       104
Denim Jeans
          2       1499         30
```

This displays products belonging to category 101.

3. Retrieve Customer Information

Display all customers

```
SQL> SELECT *
  2  FROM SHOPSPHERE_CUSTOMER;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE           CITY
--------------- --------------------------------------------------
        101 Arun
arun@gmail.com
9876543210      Chennai

        102 Priya
priya@gmail.com
9876543211      Madurai

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE           CITY
--------------- --------------------------------------------------

        103 Kumar
kumar@gmail.com
9876543212      Coimbatore
```

Search for a particular customer

```
SQL> SELECT *
  2  FROM SAVANA_CUSTOMER
  3  WHERE CUSTOMER_ID = 101;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE           CITY
--------------- --------------------------------------------------
        101 Arun
arun@gmail.com
9876543210      Chennai
```

4.Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL> SELECT *
  2  FROM SAVANA_CUSTOMER
  3  WHERE CUSTOMER_ID = 101;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE           CITY
--------------- --------------------------------------------------
        101 Arun
arun@gmail.com
9876543210      Chennai


SQL> SELECT *
  2  FROM SHOP_PRODUCT
  3  WHERE PRICE < 1000
  4  AND STOCK > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       107
Casual Shirt
        102        799         35

       108
Face Wash
        104        499         15

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

       901
Pen
          2        150         50

       902
Book
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
          2        250         30

       903
Pencil
          2        100         60

       103

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
Cotton Shirt
          2        899         40


6 rows selected.
```

4.2 OR Condition

```
SQL> SELECT *
  2  FROM SHOP_PRODUCT
  3  WHERE CATEGORY_ID = 1
  4  OR CATEGORY_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       101
Wireless Headphones
          1       1799         30

       901
Pen
          2        150         50

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

       902
Book
          2        250         30

       903
Pencil

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
          2        100         60

       102
Smart Watch
          1       2999         15

       103
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
          2        100         60

       102
Smart Watch
          1       2999         15

       103

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
Cotton Shirt
          2        899         40

       104
Denim Jeans
          2       1499         30


7 rows selected.
```

This displays products belonging to either category 101 or 102.

4.3 IN Operator

```
SQL> SELECT *
  2  FROM SHOP_PRODUCT
  3  WHERE CATEGORY_ID IN (1, 2, 3);
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       101
Wireless Headphones
          1       1799         30

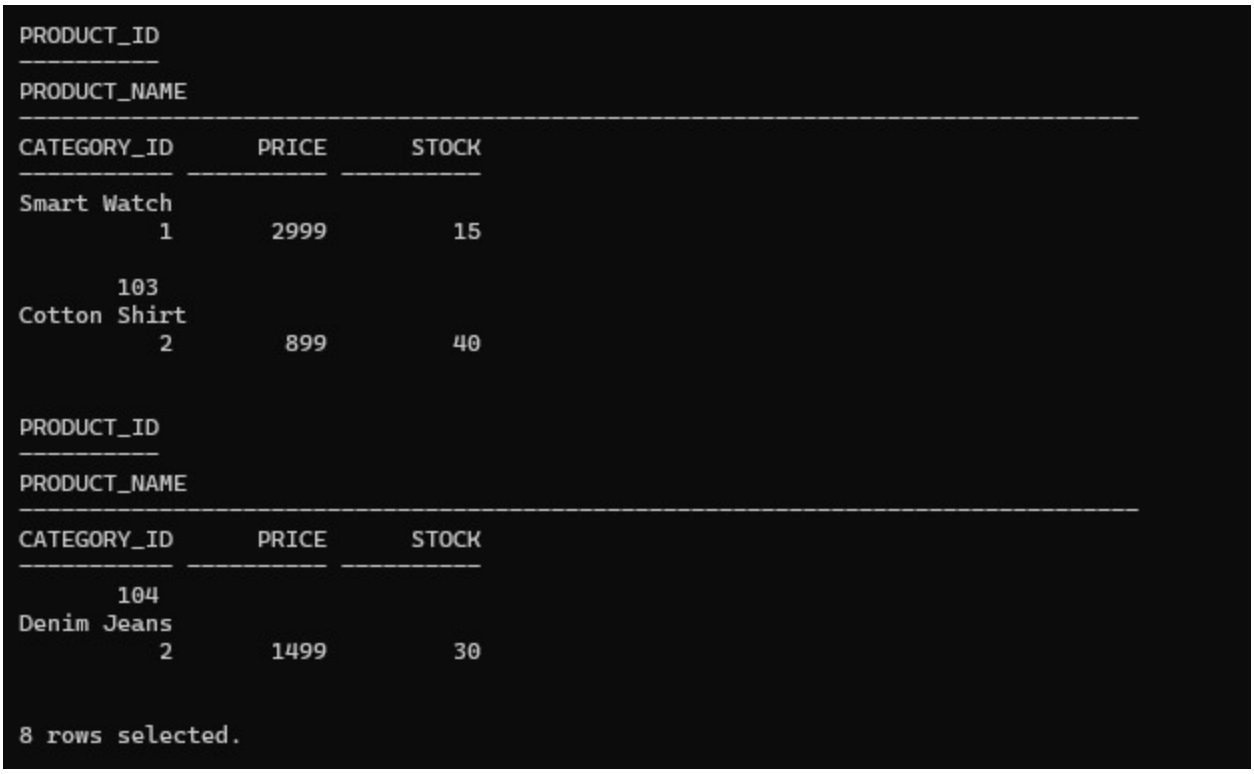
       105
Electric Kettle
          3       1299         20
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

       901
Pen
          2        150         50

       902
Book
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
          2        250         30

       903
Pencil
          2        100         60

       102
```



4.4 BETWEEN Operator

```
SQL> SELECT *
  2  FROM SHOP_PRODUCT
  3  WHERE PRICE BETWEEN 500 AND 2000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       101
Wireless Headphones
          1       1799         30

       105
Electric Kettle
          3       1299         20

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------

       106
Bluetooth Speaker
        101       1599         20

       107
Casual Shirt

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
        102        799         35

       109
Handbag
        102       1299         10
```

```
       109
Handbag
        102       1299         10

       110

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
Electric Kettle
        103       1199         18

       103
Cotton Shirt
          2        899         40


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE      STOCK
----------- ---------- ----------
       104
Denim Jeans
          2       1499         30


8 rows selected.
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM SHOP_PRODUCT
  3  ORDER BY PRICE DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       102
Smart Watch
      2999

       101
Wireless Headphones
      1799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       106
Bluetooth Speaker
      1599

       104
Denim Jeans

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1499

       105
Electric Kettle
      1299
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       108
Face Wash
       499

       902
Book

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       250

       901
Pen
       150

       903

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Pencil
       100


13 rows selected.
```

5.2 Customer Order Report

```
SQL> SELECT
  2      ORDER_ID,
  3      CUSTOMER_ID,
  4      ORDER_DATE,
  5      TOTAL_AMOUNT,
  6      ORDER_STATUS
  7  FROM SAVANA_ORDERS;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----------- --------- ------------ ------------------------------
      1001         101 22-SEP-26        26998 Confirmed
      1002         102 21-SEP-26         1999 Shipped
      1003         103 20-SEP-26         1598 Delivered
```

5.3 Customer-wise Total Orders

```
SQL> SELECT
  2      CUSTOMER_ID,
  3      COUNT(ORDER_ID) AS TOTAL_ORDERS
  4  FROM SAVANA_ORDERS
  5  GROUP BY CUSTOMER_ID;

CUSTOMER_ID TOTAL_ORDERS
----------- ------------
        101            2
        102            1
        103            1
```

5.4 Customer-wise Total Purchase Amount

```
SQL> SELECT
  2      CUSTOMER_ID,
  3      SUM(TOTAL_AMOUNT) AS TOTAL_PURCHASE_AMOUNT
  4  FROM SAVANA_ORDERS
  5  GROUP BY CUSTOMER_ID;

CUSTOMER_ID TOTAL_PURCHASE_AMOUNT
----------- ---------------------
        101                 42997
        102                  1999
        103                  1598
```